MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced schottky TTL, bus transceivers, with three-state outputs. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quad inverting bus transceivers, with three-state outputs
02	Quad noninverting bus transceivers, with three-state outputs
03	Octal noninverting bus transceivers, with three-state outputs

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
R	D-8 (20-lead, 1/4" x 1 1/16"), dual-in-line package
S	F-9 (20-lead, 1/4" x 1/2"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-01-21 14:25Z
Check the source to verify that this is the current version before use.

1.3 Absolute maximum ratings.

Supply voltage range	-.5 V to +7.0 V
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	
Device type 01	495 mW
Device type 02	495 mW
Device type 03	605 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC})	
Cases A, B, C, D, R, and S	(see MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high-level input voltage (V_{IH})	2.0 V
Maximum low-level input voltage (V_{IL})	0.8 V
Normalized fanout (each output)	
Low-logic level	33 maximum
High-logic level	50 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.

3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V; I_{OH} = -3 mA V_{IL} = .8 V; V_{IH} = 2.0 V	All	2.4		V
	V_{OH1}	V_{CC} = 4.5 V; I_{OH} = -12 mA V_{IL} = .8 V; V_{IH} = 2.0 V	All	2.0 V		"
	V_{OH2}	V_{CC} = 4.5 V; I_{OH} = -1 mA V_{IL} = .8 V; V_{IH} = 2.0 V	03	2.5		"
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V; I_{OL} = 48 mA V_{IL} = .8 V; V_{IH} = 2.0 V	All		.55	"
	V_{OL1}	V_{CC} = 4.5 V; I_{OL} 20 mA V_{IL} = .8 V; V_{IH} = 2.0 V	03		.5	"
Input clamp voltage	V_{IC}	I_I = -18 mA; T_C = +25°C V_{CC} = 0.0 V	All		-1.2	"
High-level input current	I_{IH1}	V_{CC} = 5.5 V; V_{IH} = 2.7 V	All		20	µA
	I_{IH2}	V_{CC} = 5.5 V; V_{IH} = 7.0 V	All		100	
	I_{IH3}	V_{CC} = 5.5 V; V_{IH} = 5.5 V	All		1.0	mA
Low-level input current	I_{IL1}	V_{CC} = 5.5 V V_{IL} = .5 V	01	-.03	-1.0	mA
			02	-.03	-1.0	"
			03	-.04	-1.6	"
Low-level input current	I_{IL2}	V_{CC} = 5.5 V V_{IL} = .5 V	02	-.06	-1.6	mA
			03	-.04	-1.2	"
Short current output current 1/	I_{OS}	V_{CC} = 5.5 V; V_{OS} = 0 V	All	-100	-325	mA
	I_{OS1}	V_{CC} = 5.5 V; V_{OS} = 0 V	03	-60	-150	"
Low-level input, off-state low-output current	I_{IOZL}	V_{CC} = 5.5 V V_{IOZL} = 0.5 V	01	-.42	-1.0	mA
			02	-.03	-1.6	"
			03	-.03	-1.6	"
High-level input, off-state high-output current	I_{IOZH}	V_{CC} = 5.5 V V_{IOZH} = 2.7 V	01		70	µA
			02		"	"
			03		"	"
Supply current	I_{CCH}	V_{CC} = 5.5 V; V_{IN} = 0.0 V	01		60	mA
		V_{CC} = 5.5 V; V_{IN} = 5.5 V	02		80	"
			03		110	"
Supply current	I_{CCL}	V_{CC} = 5.5 V; V_{IN} = 5.5 V	01		90	mA
		V_{CC} = 5.5 V; V_{IN} = 0.0 V	02		92	"
			03		130	"
Supply current	I_{CCZ}	V_{CC} = 5.5 V V_{IN} = OPEN	01		90	mA
			02		90	"
			03		143	"

1/ Not more than one output should be shorted at a time.

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
Propagation delay time low-to-high level		V_{CC} = 5.0 V (See figure 4)				
Bn to An	t_{PLH1}		01	1.0	6.5	ns
			02	1.0	6.5	"
			03	1.0	8.0	"
An to Bn	t_{PLH2}		01	1.0	6.5	ns
			02	1.0	6.5	"
			03	1.0	8.5	"
Propagation delay time high-to-low level		V_{CC} = 5.0 V (See figure 4)				
Bn to An	t_{PHL1}		01	1.0	8.5	ns
			02	1.0	8.5	"
			03	1.0	8.0	"
An to Bn	t_{PHL2}		01	1.0	8.5	ns
			02	1.0	8.5	"
			03	1.0	8.5	"
Propagation delay time low-level-to off-state		V_{CC} = 5.0 V (See figure 4)				
E, $\overline{E}$, $\overline{OE}$ to An	t_{PLZ1}		01	2.0	8.5	ns
			02	"	8.5	"
			03	"	10.0	"
E, $\overline{E}$, $\overline{OE}$ to Bn	t_{PLZ2}		01	"	8.5	ns
			02	"	8.5	"
			03	"	10.0	"
Propagation delay time high-level to off-state		V_{CC} = 5.0 V (See figure 4)				
E, $\overline{E}$, $\overline{OE}$ to An	t_{PHZ1}		01	2.0	7.5	ns
			02	"	7.5	"
			03	"	9.0	"
E, $\overline{E}$, $\overline{OE}$ to Bn	t_{PHZ2}		01	"	7.5	ns
			02	"	7.5	"
			03	"	9.0	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
Propagation delay off-state to low-level		V_{CC} = 5.0 V (See figure 4)				
E, $\overline{E}$, $\overline{OE}$ to An	t_{PZL1}		01	2.0	10.5	ns
			02	"	10.5	"
			03	"	13.0	"
E, $\overline{E}$, $\overline{OE}$ to Bn	t_{PZL2}		01	"	10.5	ns
			02	"	10.5	"
			03	"	13.0	"
Propagation delay off-state to high-level		V_{CC} = 5.0 V (See figure 4)				
E, $\overline{E}$, $\overline{OE}$ to An	t_{PZH1}		01	2.0	8.0	ns
			02	"	8.0	"
			03	"	9.5	"
E, $\overline{E}$, $\overline{OE}$ to Bn	t_{PZH2}		01	"	8.0	ns
			02	"	8.0	"
			03	"	9.5	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,7,8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1, 2, 3
Additional electrical subgroups for group C periodic inspections	N/A	10, 11
Group D end-point electrical parameters (method 5005)	1, 2, 3	1, 2, 3

*PDA applies to subgroup 1 (see 4.2c).

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:25Z
Check the source to verify that this is the current version before use.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 9 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Test requirements shall be as specified in table II, herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices and shall consist of the tests, conditions and limits specified for subgroups 10 and 11 of group A.

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

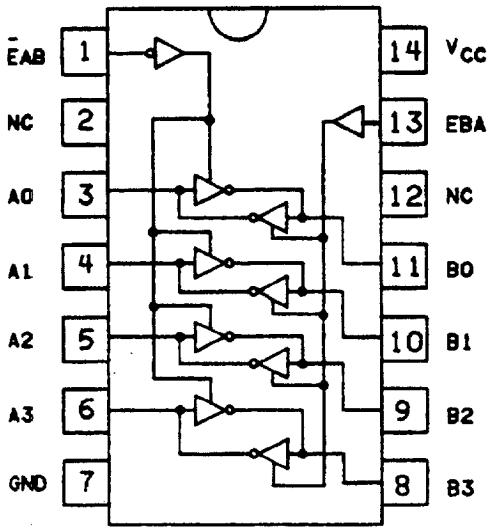
1. Test condition D or E, using the circuit shown on figure 3, or equivalent.

2. T_A = +125°C, minimum.

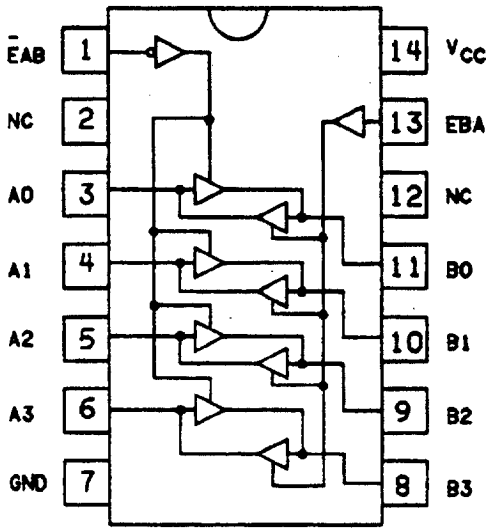
3. Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

Source: https://assist.dla.mil -- Downloaded: 2026l0-21T14:25Z
Check the source to verify that this is the currentversion before use.

Device type 01
Cases A, B, C and D



Device type 02
Cases A, B, C and D



Device type 03
Cases R and S

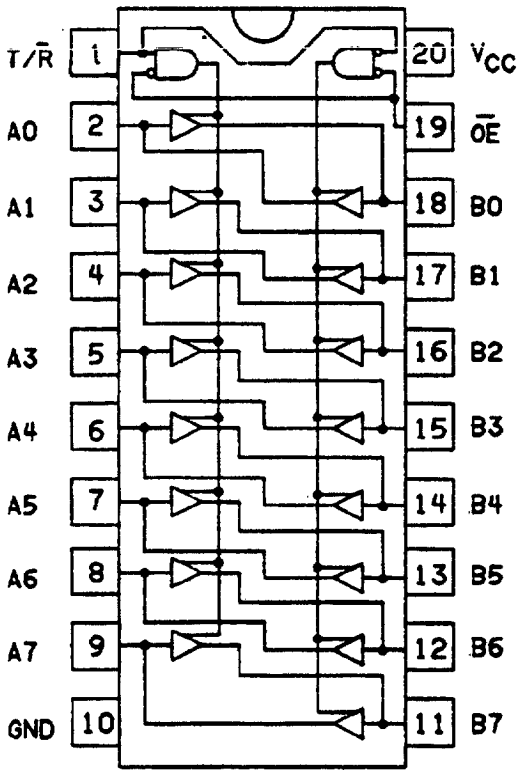
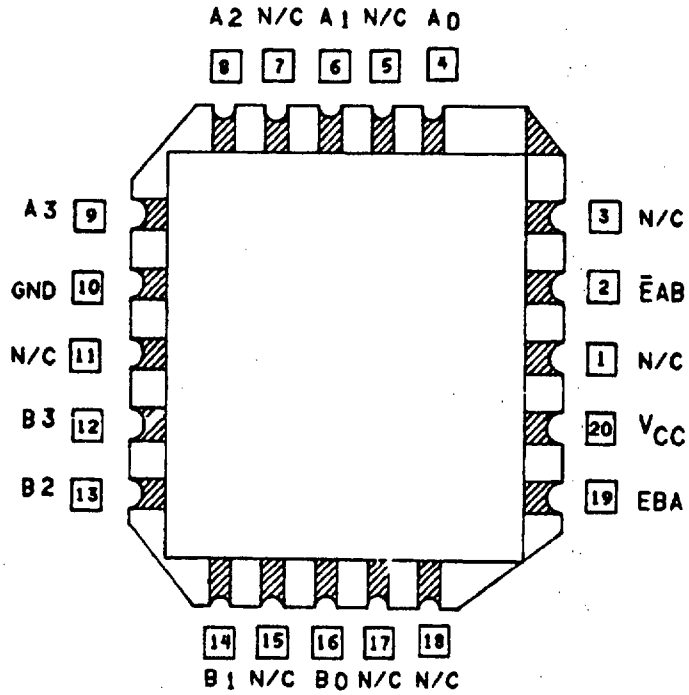


FIGURE 1. Terminal connections and logic diagrams.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

Device types 01 and 02

Case 2



Device type 03

Case 2

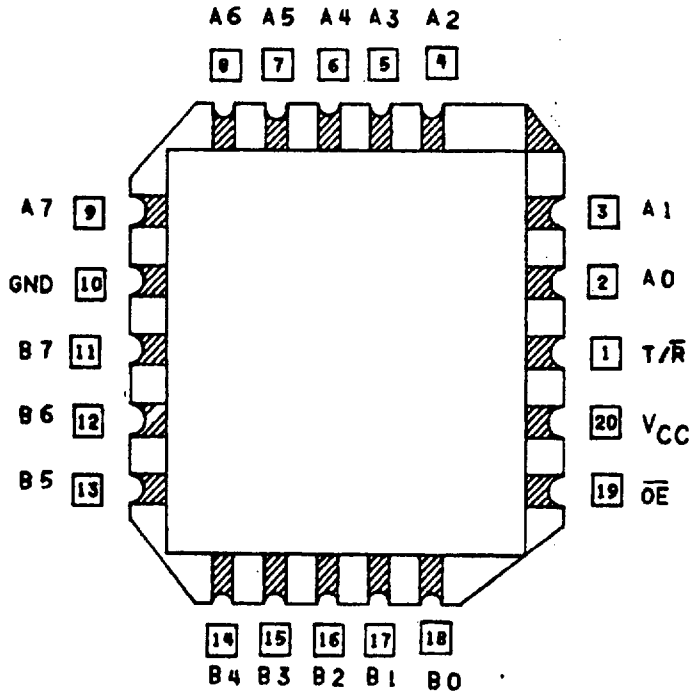


FIGURE 1. Terminal connections and logic diagrams - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

Device type 01

INPUTS				OUTPUTS	
$\overline{E}AB$	EBA	An	Bn	Bn	An
L	L	H	-	L	-
L	L	L	-	H	-
H	H	-	H	-	L
H	H	-	L	-	H
H	L	X	-	Z	-
H	L	-	X	-	Z

Device type 02

INPUTS				OUTPUTS	
$\overline{E}AB$	EBA	An	Bn	Bn	An
L	L	H	-	H	-
L	L	L	-	L	-
H	H	-	H	-	H
H	H	-	L	-	L
H	L	X	-	Z	-
H	L	-	X	-	Z

Device type 03

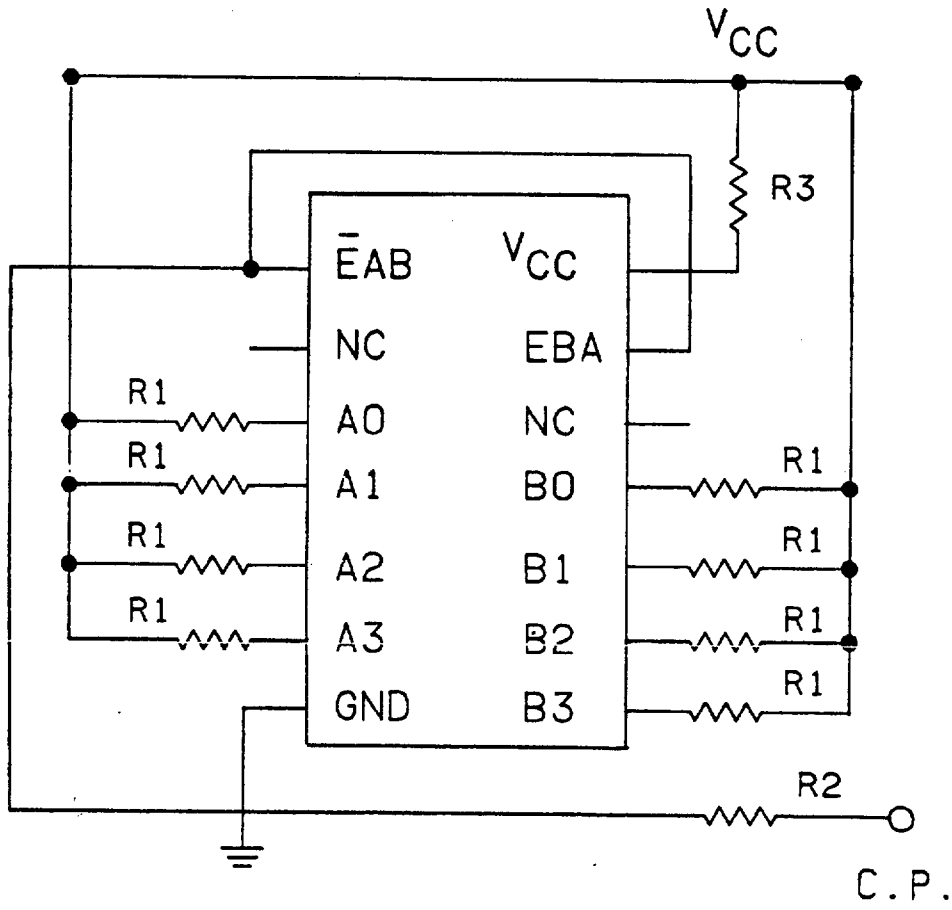
INPUTS				OUTPUTS	
$\overline{QE}$	T/$\overline{R}$	An	Bn	Bn	An
L	H	H	-	H	-
L	H	L	-	L	-
L	L	-	H	-	H
L	L	-	L	-	L
H	H	X	-	Z	-
H	L	-	X	-	Z

H = High; L = Low; X = Don't Care
Z = High Impedance

FIGURE 2. Truth tables.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

Device types 01 and 02

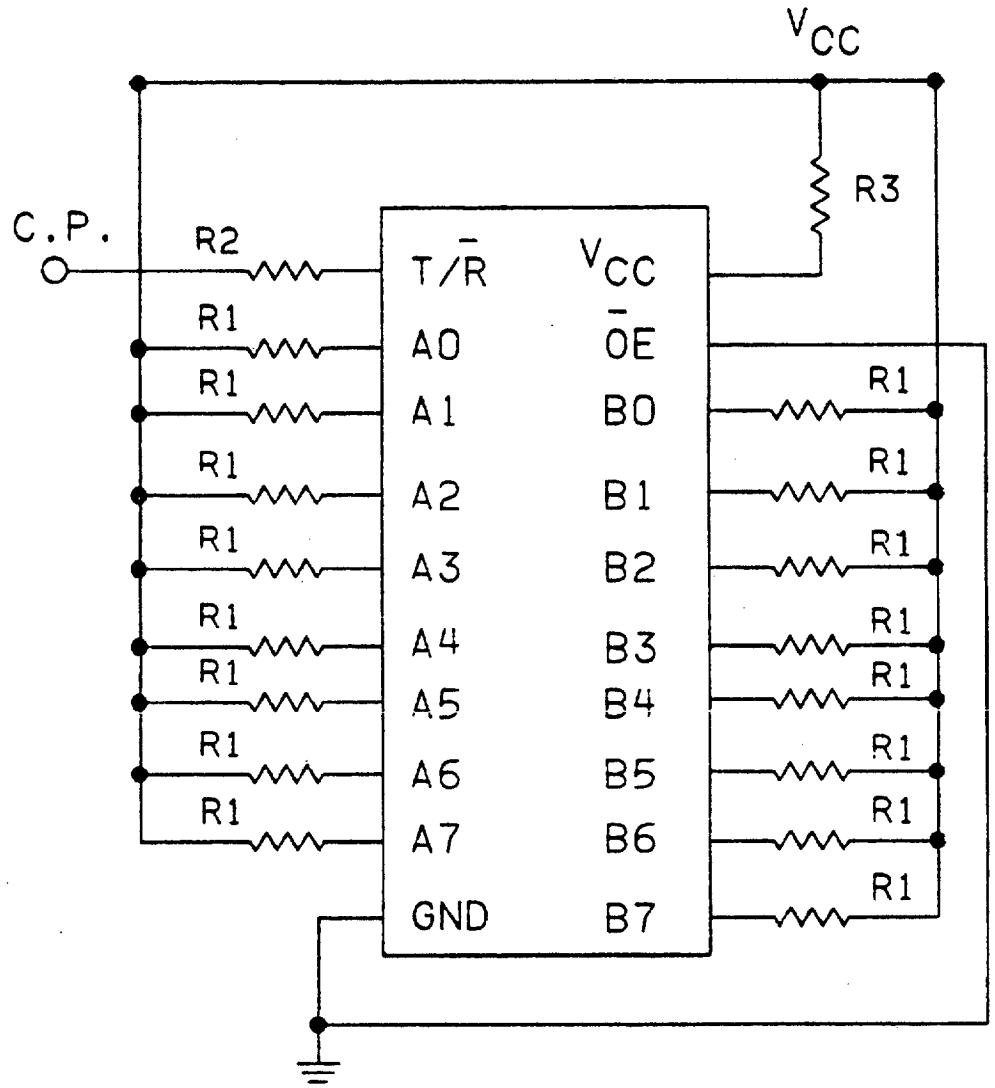


NOTES:
1. CP = 100 kHz ±50% square wave; duty cycle = 50 ±15%.
   V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%, R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

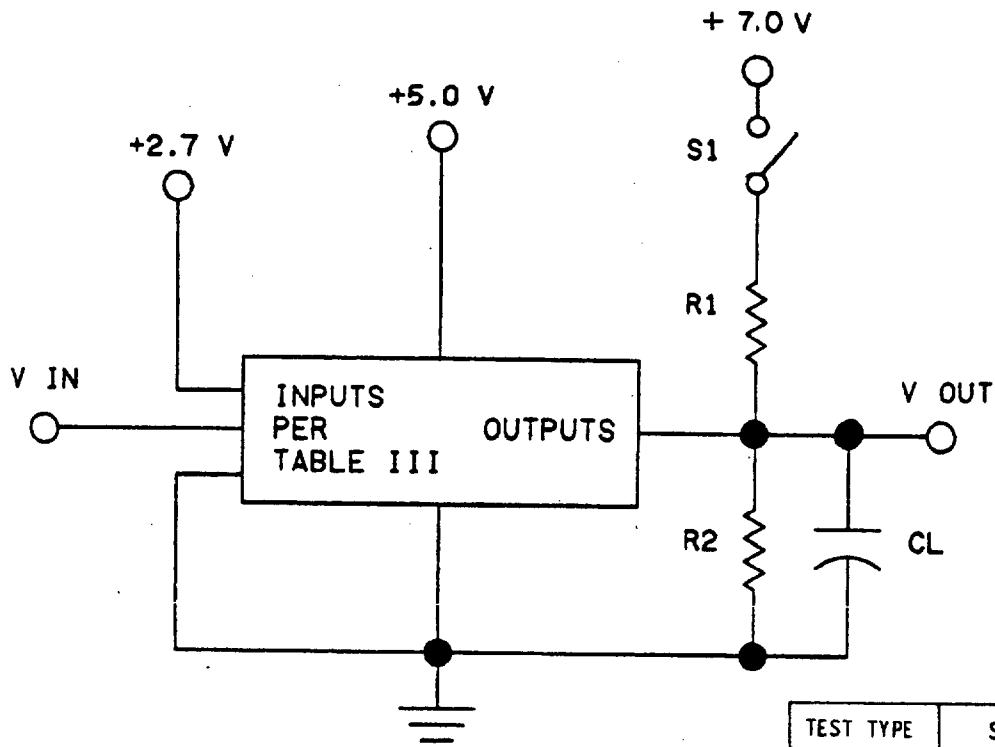
Device type 03



NOTES:
1. CP or $\overline{CP}$ = 100 kHz ±50% square wave; duty cycle = 50 ±15%;
   V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 100Ω ±5%, R2 = 51Ω ±5%.
3. V_{CC} and R3 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

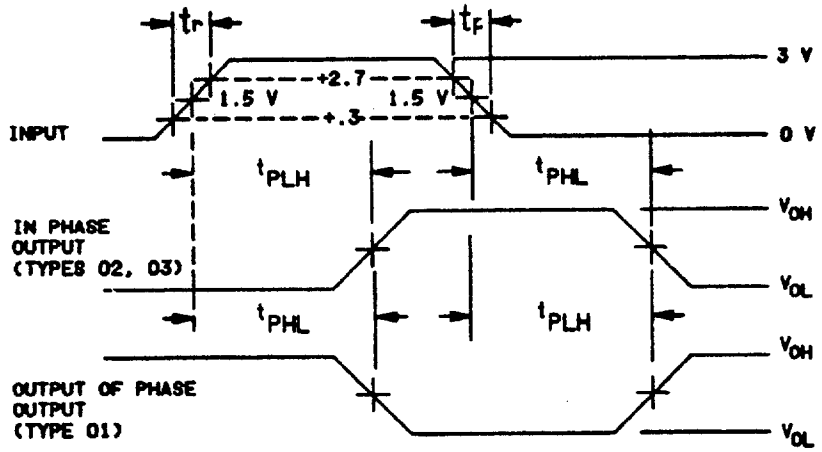


TEST TYPE	S1
t_{PLH}	OPEN
t_{PHL}	OPEN
t_{PHZ}	OPEN
t_{PZH}	OPEN
t_{PLZ}	CLOSED
t_{PZL}	CLOSED

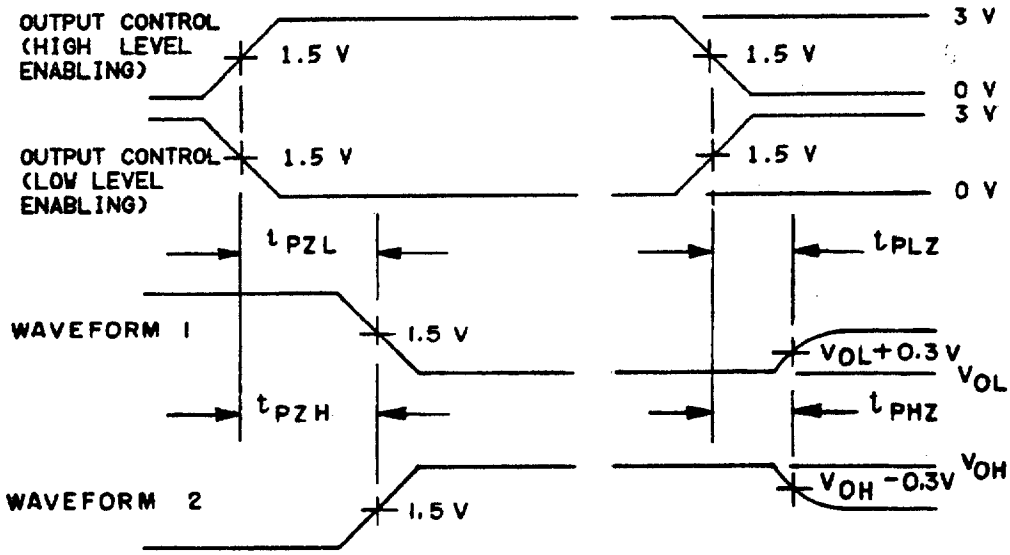
NOTES:
1. V_{IN} = input pulse and has the following characteristics: PRR = 1 MHz, $t_r = t_f \leq 2.5$ ns.
2. C_L = 50 pF ±10%, including scope probe, wiring and stray capacitance without package in test fixture.
3. Voltage measurements are to be made with respect to network ground terminal.
4. R1 = R2 = 500Ω ±5%.

FIGURE 4. Switching time test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.



VOLTAGE WAVEFORMS
PROPAGATION DELAY TIMES



VOLTAGE WAVEFORMS
ENABLE AND DISABLE TIMES, THREE-STATE OUTPUTS

FIGURE 4. Switching time waveforms - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{E}_{AB}$	NC	A0	A1	A2	A3	GND	B3	B2	B1	B0	NC	$\overline{E}_{BA}$	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V		-3 mA	-3 mA	-3 mA	-3 mA	GND	0.8 V	0.8 V	0.8 V	0.8 V		2.0 V	4.5 V	A0	2.4		V
		"	2	"					0.8 V	"	-3 mA					"	"	A1	"		"
		"	3	"						"		-3 mA	-3 mA			"	"	A2	"		"
		"	4	"						"				-3 mA		"	"	A3	"		"
		"	5	0.8 V						"						0.8 V	"	B3	"		"
		"	6	"						"						"	"	B2	"		"
		"	7	"			0.8 V			"						"	"	B1	"		"
		"	8	"		0.8 V				"						"	"	B0	"		"
	V_{OH1}	3006	9	2.0 V		-12 mA	-12 mA	-12 mA	-12 mA	"	0.8 V	0.8 V	0.8 V	0.8 V		2.0 V	"	A0	2.0		"
		"	10	"					0.8 V	"	-12 mA					"	"	A1	"		"
		"	11	"						"		-12 mA	-12 mA			"	"	A2	"		"
		"	12	"						"				-12 mA		"	"	A3	"		"
		"	13	0.8 V						"						0.8 V	"	B3	"		"
		"	14	"						"						"	"	B2	"		"
		"	15	"			0.8 V			"						"	"	B1	"		"
		"	16	"		0.8 V				"						"	"	B0	"		"
	V_{OL}	3007	17	2.0 V		48 mA	48 mA	48 mA	48 mA	"	2.0 V	2.0 V	2.0 V	2.0 V		2.0 V	"	A0		.55	"
		"	18	"					2.0 V	"	48 mA					"	"	A1		"	"
		"	19	"						"		48 mA	48 mA			"	"	A2		"	"
		"	20	"						"				48 mA		"	"	A3		"	"
		"	21	0.8 V						"						0.8 V	"	B3		"	"
		"	22	"						"						"	"	B2		"	"
		"	23	"			2.0 V			"						"	"	B1		"	"
		"	24	"		2.0 V				"						"	"	B0		"	"
	V_{IC}		25	-18 mA		-18 mA	-18 mA	-18 mA	-18 mA	"	-18 mA	-18 mA	-18 mA	-18 mA		0.0 V	"	$\overline{E}_{AB}$		-1.2	"
			26	5.5 V						"						"	"	A0		"	"
			27	"						"						"	"	A1		"	"
			28	"						"						"	"	A2		"	"
			29	"						"						"	"	A3		"	"
			30	"						"						"	"	B3		"	"
			31	"						"						"	"	B2		"	"
			32	"						"						"	"	B1		"	"
			33	"						"						"	"	B0		"	"
			34	"						"						-18 mA	"	$\overline{E}_{BA}$		"	"
	I_{IH1}	3010	35	2.7 V						"						2.7 V	5.5 V	$\overline{E}_{AB}$		20	µA
		"	36	"						"						"	"	$\overline{E}_{BA}$		"	"
	I_{IH2}	"	37	7.0 V						"						7.0 V	"	$\overline{E}_{AB}$		100	"
		"	38	"						"						"	"	$\overline{E}_{BA}$		"	"

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	13	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{EAB}$	NC	A0	A1	A2	A3	GND	B3	B2	B1	B0	NC	EBA	V_{CC}				
1 T_C = +25°C	I_{IH3}	3010	39	0.0 V		5.5 V				GND						0.0 V	5.5 V	A0		1.0	mA
			40	"			5.5 V			"						"	"	A1		"	"
			41	"				5.5 V		"						"	"	A2		"	"
			42	"					5.5 V	"						"	"	A3		"	"
			43	5.5 V						"	5.5 V					5.5 V	"	B3		"	"
			44	"						"		5.5 V				"	"	B2		"	"
			45	"						"			5.5 V			"	"	B1		"	"
			46	"						"				5.5 V		"	"	B0		"	"
	I_{IL1}	3009	47	0.5						"						0.5 V	"	$\overline{EAB}$ EBA	1/ "	1/ "	" "
			48																		
	I_{OS}	3011	49	"		0.0 V				"				0.0 V		5.5 V	"	A0	-100	-325	"
			50	"			0.0 V			"			0.0 V			"	"	A1	"	"	"
			51	"				0.0 V		"		0.0 V				"	"	A2	"	"	"
			52	"					0.0 V	"	0.0 V					"	"	A3	"	"	"
			53	0.0 V					0.0 V	"	0.0 V					0.0 V	"	B3	"	"	"
			54	"				0.0 V		"		0.0 V				"	"	B2	"	"	"
			55	"			0.0 V			"			0.0 V			"	"	B1	"	"	"
			56	"		0.0 V				"				0.0 V		"	"	B0	"	"	"
	I_{IOZL}		57	2.0 V		0.5 V				"				0.0 V		0.8 V	"	A0	1/	1/	"
			58	"			0.5 V			"			0.0 V			"	"	A1	"	"	"
			59	"				0.5 V		"		0.0 V				"	"	A2	"	"	"
			60	"					0.5 V	"	0.0 V					"	"	A3	"	"	"
			61	"					0.0 V	"	0.5 V					"	"	B3	"	"	"
			62	"				0.0 V		"		0.5 V				"	"	B2	"	"	"
			63	"			0.0 V			"			0.5 V			"	"	B1	"	"	"
			64	"		0.0 V				"				0.5 V		"	"	B0	"	"	"
	I_{IOZH}		65	"		2.7 V				"				5.5 V		0.8 V	5.5 V	A0		70	µA
			66	"			2.7 V			"			5.5 V			"	"	A1		"	"
			67	"				2.7 V		"		5.5 V				"	"	A2		"	"
			68	"					2.7 V	"	5.5 V					"	"	A3		"	"
			69	"					5.5 V	"	2.7 V					"	"	B3		"	"
			70	"				5.5 V		"		2.7 V				"	"	B2		"	"
			71	"			5.5 V			"			2.7 V			"	"	B1		"	"
			72	"		5.5 V				"				2.7 V		"	"	B0		"	"

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
				2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	EAB	NC	A0	A1	A2	A3	GND	B3	B2	B1	B0	NC	EBA	V_{CC}				
1 T_C = +25°C	I_{CCH}	3005	73	5.5 V						GND	0.0 V	0.0 V	0.0 V	0.0 V		5.5 V	5.5 V	V_{CC}		60	mA
		"	74	0.0 V		0.0 V	0.0 V	0.0 V	0.0 V	"						0.0 V	"			60	"
	I_{CCL}	"	75	5.5 V						"	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V	"	"		90	"
		"	76	0.0 V		5.5 V	5.5 V	5.5 V	5.5 V	"						0.0 V	"	"		"	"
	I_{CCZ}	"	77	5.5 V						"						0.0 V	"	"		"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional tests 2/	3014	78	A		L	L	L	L	GND	A	A	A	A		A	3/	All outputs			
		"	79	A		H	H	H	H	"	B	B	B	B		A	"	"			
		"	80	B		A	A	A	A	"	L	L	L	L		B	"	"			
			81	B		B	B	B	B	"	H	H	H	H		B	"	"			
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003	82	2.7 V		OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN		2.7 V	5.0 V	B0 to A0	1.5	5.2	ns
		"	83	"						"						"	"	B1 to A1	"	"	"
		"	84	"						"						"	"	B2 to A2	"	"	"
		"	85	"						"						"	"	B3 to A3	"	"	"
	t_{PLH2}	"	86	0.0 V		IN	IN	IN	IN	"	OUT	OUT	OUT	OUT		0.0 V	"	A0 to B0	"	"	"
		"	87	"						"						"	"	A1 to B1	"	"	"
		"	88	"						"						"	"	A2 to B2	"	"	"
		"	89	"						"						"	"	A3 to B3	"	"	"
	t_{PHL1}	"	90	2.7 V		OUT	OUT	OUT	OUT	"	IN	IN	IN	IN		2.7 V	"	B0 to A0	"	"	"
		"	91	"						"						"	"	B1 to A1	"	"	"
		"	92	"						"						"	"	B2 to A2	"	"	"
		"	93	"						"						"	"	B3 to A3	"	"	"
	t_{PHL2}	"	94	0.0 V		IN	IN	IN	IN	"	OUT	OUT	OUT	OUT		0.0 V	"	A0 to B0	"	"	"
		"	95	"						"						"	"	A1 to B1	"	"	"
		"	96	"						"						"	"	A2 to B2	"	"	"
		"	97	"						"						"	"	A3 to B3	"	"	"
	t_{PLZ1}	"	98	2.7 V		OUT	OUT	OUT	OUT	"	2.7 V	2.7 V	2.7 V	2.7 V		IN	"	EBA to A0	2.5	6.0	"
		"	99	"						"						"	"	EBA to A1	"	"	"
		"	100	"						"						"	"	EBA to A2	"	"	"
		"	101	"						"						"	"	EBA to A3	"	"	"

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{EAB}$	NC	AO	A1	A2	A3	GND	B3	B2	B1	B0	NC	$\overline{EBA}$	V_{CC}				
9 T_C = +25°C	t_{PLZ2}	3003	102	IN		2.7 V	2.7 V	2.7 V	2.7 V	GND		OUT	OUT	OUT		0.0 V	5.0 V	$\overline{EAB}$ to B0	2.5	6.0	ns
		"	103	"		"				"						"	"	$\overline{EAB}$ to B1	"	"	"
		"	104	"		"				"						"	"	$\overline{EAB}$ to B2	"	"	"
		"	105	"		"				"	OUT					"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PHZ1}	"	106	2.7 V		OUT	OUT	OUT	OUT	"		OUT	0.0 V	0.0 V		IN	"	$\overline{EBA}$ to A0	"	"	"
		"	107	"						"						"	"	$\overline{EBA}$ to A1	"	"	"
		"	108	"						"						"	"	$\overline{EBA}$ to A2	"	"	"
		"	109	"						"	0.0 V	0.0 V	0.0 V			"	"	$\overline{EBA}$ to A3	"	"	"
	t_{PHZ2}	"	110	IN		0.0 V	0.0 V	0.0 V	0.0 V	"		OUT	OUT	OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	111	"						"						"	"	$\overline{EAB}$ to B1	"	"	"
		"	112	"						"						"	"	$\overline{EAB}$ to B2	"	"	"
		"	113	"						"	OUT					"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PZL1}	"	114	2.7 V		OUT	OUT	OUT	OUT	"		2.7 V	2.7 V	2.7 V		IN	"	$\overline{EBA}$ to A0	"	8.5	"
		"	115	"						"						"	"	$\overline{EBA}$ to A1	"	"	"
		"	116	"						"						"	"	$\overline{EBA}$ to A2	"	"	"
		"	117	"						"	2.7 V	2.7 V	2.7 V			"	"	$\overline{EBA}$ to A3	"	"	"
	t_{PZL2}	"	118	IN		2.7 V	2.7 V	2.7 V	2.7 V	"		OUT	OUT	OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	119	"						"						"	"	$\overline{EAB}$ to B1	"	"	"
		"	120	"						"						"	"	$\overline{EAB}$ to B2	"	"	"
		"	121	"						"	OUT					"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PZH1}	"	122	2.7 V		OUT	OUT	OUT	OUT	"		OUT	0.0 V	0.0 V		IN	"	$\overline{EBA}$ to A0	"	5.7	"
		"	123	"						"						"	"	$\overline{EBA}$ to A1	"	"	"
		"	124	"						"						"	"	$\overline{EBA}$ to A2	"	"	"
		"	125	"						"	0.0 V	0.0 V	0.0 V			"	"	$\overline{EBA}$ to A3	"	"	"
	t_{PZH2}	"	126	IN		0.0 V	0.0 V	0.0 V	0.0 V	"		OUT	OUT	OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	127	"						"						"	"	$\overline{EAB}$ to B1	"	"	"
		"	128	"						"						"	"	$\overline{EAB}$ to B2	"	"	"
		"	129	"						"	OUT					"	"	$\overline{EAB}$ to B3	"	"	"

See footnotes at end of device type 01.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{E}AB$	NC	AO	A1	A2	A3	GND	B3	B2	B1	BO	NC	$\overline{E}BA$	V_{CC}				
10	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																	1.0	6.5	ns
	t_{PLH2}																		"	6.5	"
	t_{PHL1}																		"	8.5	"
	t_{PHL2}																		"	"	"
	t_{PLZ1}																		2.0	"	"
	t_{PLZ2}																		"	"	"
	t_{PHZ1}																		"	7.5	"
	t_{PHZ2}																		"	7.5	"
	t_{PZL1}																		"	10.5	"
	t_{PZL2}																		"	10.5	"
	t_{PZH1}																		"	8.0	"
	t_{PZH2}																		"	8.0	"
11		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																			

1/ I_{IOZL}, I_{IL1} limits shall be as follows:

Test	Min/Max limits in (mA) for circuit		
	A	B	C
	Fairchild	Motorola	Signetics
I_{IL1}	-.42/-1.0	-.03/-1.0	-.03/-1.0
I_{IOZL}	-.42/-1.0	-.03/-1.0	-.03/-1.0

2/ A = 3.0 V minimum, B = 0.0 V or GND;
H > 1.5 V, L < 1.5 V

3/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
				2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{EAB}$	NC	A0	A1	A2	A3	GND	B3	B2	B1	B0	NC	$\overline{EBA}$	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V		-3 mA	-3 mA	-3 mA		GND						2.0 V	4.5 V	A0	2.4		V
		"	2	"		"			-3 mA	"		2.0 V	2.0 V	2.0 V		"	"	A1	"		"
		"	3	"		"			2.0 V	"	2.0 V	-3 mA				"	"	A2	"		"
		"	4	"		"				"						"	"	A3	"		"
		"	5	0.8 V		"				"						0.8 V	"	B3	"		"
		"	6	"		"				"						"	"	B2	"		"
		"	7	"		"	2.0 V	2.0 V		"		-3 mA	-3 mA	-3 mA		"	"	B1	"		"
		"	8	"		2.0 V				"						"	"	B0	"		"
	V_{OL}	3007	9	2.0 V		48 mA	48 mA	48 mA		"				0.8 V		2.0 V	"	A0		.55	"
		"	10	"		"			48 mA	"		0.8 V	0.8 V			"	"	A1		"	"
		"	11	"		"			0.8 V	"	0.8 V	48 mA				"	"	A2		"	"
		"	12	"		"				"						"	"	A3		"	"
		"	13	0.8 V		"				"						0.8 V	"	B3		"	"
		"	14	"		"				"						"	"	B2		"	"
		"	15	"		"	0.8 V	0.8 V		"		48 mA	48 mA	48 mA		"	"	B1		"	"
		"	16	"		0.8 V				"						"	"	B0		"	"
	V_{OH1}	3006	17	2.0 V		-12 mA	-12 mA	-12 mA		"				2.0 V		2.0 V	"	A0	2.0		"
		"	18	"		"			-12 mA	"		2.0 V	2.0 V			"	"	A1	"		"
		"	19	"		"			2.0 V	"	2.0 V	-12 mA				"	"	A2	"		"
		"	20	"		"				"						"	"	A3	"		"
		"	21	0.8 V		"				"						0.8 V	"	B3	"		"
		"	22	"		"				"						"	"	B2	"		"
		"	23	"		"	2.0 V	2.0 V		"		-12 mA	-12 mA	-12 mA		"	"	B1	"		"
		"	24	"		2.0 V				"						"	"	B0	"		"
	V_{IC}		25	-18 mA		-18 mA	-18 mA	-18 mA		"						0.0 V	"	$\overline{EAB}$		-1.2	"
			26	5.5 V		"			-18 mA	"						"	"	A0		"	"
			27	"		"				"						"	"	A1		"	"
			28	"		"				"						"	"	A2		"	"
			29	"		"				"						"	"	A3		"	"
			30	"		"				"						"	"	B3		"	"
			31	"		"				"						"	"	B2		"	"
			32	"		"				"						"	"	B1		"	"
			33	"		"				"						"	"	B0		"	"
			34	"		"				"						-18 mA	"	EBA		"	"
	I_{IH1}	3010	35	2.7 V						"						2.7 V	5.5 V	$\overline{EAB}$		20	µA
		"	36	"						"						"	"	EBA		"	"
	I_{IH2}	"	37	7.0 V						"						7.0 V	"	$\overline{EAB}$		100	"
		"	38	"						"						"	"	EBA		"	"

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D / Case 2 / Test no.	1 / 2 / EAB	2 / 3 / NC	3 / 4 / A0	4 / 6 / A1	5 / 8 / A2	6 / 9 / A3	7 / 10 / GND	8 / 12 / B3	9 / 13 / B2	10 / 14 / B1	11 / 16 / B0	12 / 18 / NC	13 / 19 / EBA	14 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
T_C = $^{1}_{+25°C}$	I_{IH3}	3010	39	0.0 V		5.5 V				GND						0.0 V	5.5 V	A0		1.0	mA
		"	40	"			5.5 V			"						"	"	A1		"	"
		"	41	"				5.5 V		"						"	"	A2		"	"
		"	42	"					5.5 V	"						"	"	A3		"	"
		"	43	5.5 V						"	5.5 V					5.5 V	"	B3		"	"
		"	44	"						"		5.5 V				"	"	B2		"	"
		"	45	"						"			5.5 V			"	"	B1		"	"
		"	46	"						"				5.5 V		"	"	B0		"	"
	I_{IL1}	3009	47	0.5 V						"							"	$\overline{EAB}$	1/	1/	"
		"	48							"						0.5 V	"	EBA	1/	1/	"
	I_{OS}	3011	49	5.5 V		0.0 V				"				5.5 V		5.5 V	"	A0	-100	-325	"
		"	50	"			0.0 V			"			5.5 V			"	"	A1	"	"	"
		"	51	"				0.0 V		"		5.5 V				"	"	A2	"	"	"
		"	52	"					0.0 V	"	5.5 V					"	"	A3	"	"	"
		"	53	0.0 V					5.5 V	"	0.0 V					0.0 V	"	B3	"	"	"
		"	54	"				5.5 V		"		0.0 V				"	"	B2	"	"	"
		"	55	"			5.5 V			"			0.0 V			"	"	B1	"	"	"
		"	56	"		5.5 V				"				0.0 V		"	"	B0	"	"	"
	I_{IOZL}		57	2.0 V		0.5 V				"				5.5 V		0.8 V	"	A0	1/	1/	mA
			58	"			0.5 V			"			5.5 V			"	"	A1	"	"	"
			59	"				0.5 V		"		5.5 V				"	"	A2	"	"	"
			60	"					0.5 V	"	5.5 V					"	"	A3	"	"	"
			61	"					5.5 V	"	0.5 V					"	"	B3	"	"	"
			62	"				5.5 V		"		0.5 V				"	"	B2	"	"	"
			63	"			5.5 V			"			0.5 V			"	"	B1	"	"	"
			64	"		5.5 V				"				0.5 V		"	"	B0	"	"	"
	I_{IOZH}		65	"		2.7 V				"				0.0 V		0.8 V	5.5 V	A0		70	µA
			66	"			2.7 V			"			0.0 V			"	"	A1		"	"
			67	"				2.7 V		"		0.0 V				"	"	A2		"	"
			68	"					2.7 V	"	0.0 V					"	"	A3		"	"
			69	"					0.0 V	"	2.7 V					"	"	B3		"	"
			70	"				0.0 V		"		2.7 V				"	"	B2		"	"
			71	"			0.0 V			"			2.7 V			"	"	B1		"	"
			72	"		0.0 V				"				2.7 V		"	"	B0		"	"

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026102-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 2 Test no.	1 2 EAB	2 3 NC	3 4 A0	4 6 A1	5 8 A2	6 9 A3	7 10 GND	8 12 B3	9 13 B2	10 14 B1	11 16 B0	12 18 NC	13 19 EBA	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	I_{CCH}	3005 "	73 74	5.5 V 0.0 V		5.5 V	5.5 V	5.5 V	5.5 V	GND "	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V 0.0 V	5.5 V "	V_{CC} "		80 80	mA "
	I_{CCL}	" "	75 76	5.5 V 0.0 V		0.0 V	0.0 V	0.0 V	0.0 V	" "	0.0 V	0.0 V	0.0 V	0.0 V		5.5 V 0.0 V	" "	" "		92 "	" "
	I_{CCZ}	"	77	5.5 V						"						0.0 V	"	"		90	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C and V_{IC} tests are omitted.																				
7 T_C = +25°C	Functional tests 2/	3014 " "	78 79 80 81	A A B B		L H B A	L H B A	L H B A	L H B A	GND " " "	B A L H	B A L H	B A L H	B A L H		A A B B	3/ " " "	All outputs " "			
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003 " " "	82 83 84 85	2.7 V " " "		OUT	OUT	OUT	OUT	GND " " "	IN	IN	IN	IN		2.7 V " " "	5.0 V " " "	B0 to A0 B1 to A1 B2 to A2 B3 to A3	1.5 " " "	5.2 " " "	ns " " "
	t_{PLH2}	" " " "	86 87 88 89	0.0 V " " "		IN	IN	IN	IN	" " " "	OUT	OUT	OUT	OUT		0.0 V " " "	" " " "	A0 to B0 A1 to B1 A2 to B2 A3 to B3	" " " "	" " " "	" " " "
	t_{PHL1}	" " " "	90 91 92 93	2.7 V " " "		OUT	OUT	OUT	OUT	" " " "	IN	IN	IN	IN		2.7 V " " "	" " " "	B0 to A0 B1 to A1 B2 to A2 B3 to A3	" " " "	" " " "	" " " "
	t_{PHL2}	" " " "	94 95 96 97	0.0 V " " "		IN	IN	IN	IN	" " " "	OUT	OUT	OUT	OUT		0.0 V " " "	" " " "	A0 to B0 A1 to B1 A2 to B2 A3 to B3	" " " "	" " " "	" " " "
	t_{PLZ1}	" " " "	98 99 100 101	2.7 V " " "		OUT	OUT	OUT	OUT	" " " "	0.0 V	0.0 V	0.0 V	0.0 V		IN " " "	" " " "	EBA to A0 EBA to A1 EBA to A2 EBA to A3	2.5 " " "	6.0 " " "	" " " "

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{EAB}$	NC	AO	A1	A2	A3	GND	B3	B2	B1	B0	NC	$\overline{EBA}$	V_{CC}				
9 T_C = +25°C	t_{PLZ2}	3003	102	IN		0.0 V	0.0 V	0.0 V	0.0 V	GND	OUT	OUT	OUT	OUT		0.0 V	5.0 V	$\overline{EAB}$ to B0	2.5	6.0	ns
		"	103	"						"						"	"	$\overline{EAB}$ to B1	"	"	"
		"	104	"						"						"	"	$\overline{EAB}$ to B2	"	"	"
		"	105	"						"						"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PHZ1}	"	106	2.7 V		OUT	OUT	OUT	OUT	"				2.7 V		IN	"	EBA to A0	"	"	"
		"	107	"						"			2.7 V			"	"	EBA to A1	"	"	"
		"	108	"						"	2.7 V	2.7 V				"	"	EBA to A2	"	"	"
		"	109	"						"						"	"	EBA to A3	"	"	"
	t_{PHZ2}	"	110	IN		2.7 V	2.7 V	2.7 V	2.7 V	"				OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	111	"						"			OUT			"	"	$\overline{EAB}$ to B1	"	"	"
		"	112	"						"	OUT	OUT				"	"	$\overline{EAB}$ to B2	"	"	"
		"	113	"						"						"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PZL1}	"	114	2.7 V		OUT	OUT	OUT	OUT	"				0.0 V		IN	"	EBA to A0	"	7.5	"
		"	115	"						"			0.0 V			"	"	EBA to A1	"	"	"
		"	116	"						"	0.0 V	0.0 V				"	"	EBA to A2	"	"	"
		"	117	"						"						"	"	EBA to A3	"	"	"
	t_{PZL2}	"	118	IN		0.0 V	0.0 V	0.0 V	0.0 V	"				OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	119	"						"			OUT			"	"	$\overline{EAB}$ to B1	"	"	"
		"	120	"						"	OUT	OUT				"	"	$\overline{EAB}$ to B2	"	"	"
		"	121	"						"						"	"	$\overline{EAB}$ to B3	"	"	"
	t_{PZH1}	"	122	2.7 V		OUT	OUT	OUT	OUT	"				2.7 V		IN	"	EBA to A0	"	5.7	"
		"	123	"						"			2.7 V			"	"	EBA to A1	"	"	"
		"	124	"						"	2.7 V	2.7 V				"	"	EBA to A2	"	"	"
		"	125	"						"						"	"	EBA to A3	"	"	"
	t_{PZH2}	"	126	IN		2.7 V	2.7 V	2.7 V	2.7 V	"				OUT		0.0 V	"	$\overline{EAB}$ to B0	"	"	"
		"	127	"						"			OUT			"	"	$\overline{EAB}$ to B1	"	"	"
		"	128	"						"	OUT	OUT				"	"	$\overline{EAB}$ to B2	"	"	"
		"	129	"						"						"	"	$\overline{EAB}$ to B3	"	"	"

See footnotes at end of device type 02.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 02 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	$\overline{EA8}$	NC	A0	A1	A2	A3	GND	B3	B2	B1	B0	NC	EBA	V_{CC}				
10	t_{PLH1}																		1.0	6.5	ns
	t_{PLH2}																		"	6.5	"
	t_{PHL1}																		"	8.5	"
	t_{PHL2}																		"	"	"
	t_{PLZ1}																		2.0	"	"
	t_{PLZ2}																		"	"	"
	t_{PHZ1}		Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																"	7.5	"
	t_{PHZ2}																		"	7.5	"
	t_{PZL1}																		"	10.5	"
	t_{PZL2}																		"	10.5	"
	t_{PZH1}																		"	8.0	"
	t_{PZH2}																		"	8.0	"
11			Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																		

1/ I_{IOZL}, I_{IL1} limits shall be as follows:

Test	Min/Max limits in (mA) for circuit		
	A	B	C
	Fairchild	Motorola	Signetics
I_{IL1}	-.42/-1.0	-.03/-1.0	-.03/-1.0
I_{IL2}	-.67/-1.6	-.06/-1.6	-.06/-1.6
I_{IOZL}	-.42/-1.0	-.03/-1.0	-.03/-1.0

2/ A = 3.0 V minimum, B = 0.0 V or GND;
H > 1.5 V, L < 1.5 V

3/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test Limits Min	Test Limits Max	Unit
				T/R	A0	A1	A2	A3	A4	A5	A6	A7	GND	B7	B6	B5	B4	B3	B2	B1	B0	OE	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	GND	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	-3 mA	0.8 V	4.5 V	B0	2.4		V
			2																					B1			
			3																					B2			
			4																					B3			
			5																					B4			
			6																					B5			
			7																					B6			
			8																					B7			
	V_{OL}	3007	9		0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V		48 mA	48 mA	48 mA	48 mA	48 mA	48 mA	48 mA	48 mA			B0		.55	
			10																					B1			
			11																					B2			
			12																					B3			
			13																					B4			
			14																					B5			
			15																					B6			
			16																					B7			
	V_{OH1}	3006	17		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V		-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA	-12 mA			B0	2.0		
			18																					B1			
			19																					B2			
			20																					B3			
			21																					B4			
			22																					B5			
			23																					B6			
			24																					B7			
	V_{OH2}		25	0.8 V	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA	-1 mA		2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V			A0	2.5		
			26																					A1			
			27																					A2			
			28																					A3			
			29																					A4			
			30																					A5			
			31																					A6			
			32																					A7			
	V_{OL1}	3007	33		20 mA	20 mA	20 mA	20 mA	20 mA	20 mA	20 mA	20 mA		0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V			A0		0.5	
			34																					A1			
			35																					A2			
			36																					A3			
			37																					A4			
			38																					A5			
			39																					A6			
			40																					A7			
	V_{IC}		41	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA		4.5 V	T/R		-1.2	
			42																				0.0 V	A0			
			43																					A1			
			44																					A2			
			45																					A3			
			46																					A4			
			47																					A5			
			48																					A6			
			49																					A7			
			50																					B7			
			51																					B6			
			52																					B5			
			53																					B4			
			54																					B3			
			55																					B2			
			56																					B1			
			57																					B0			
			58																					OE			

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be High ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				T/R	A0	A1	A2	A3	A4	A5	A6	A7	GND	B7	B6	B5	B4	B3	B2	B1	B0	$\overline{OE}$	V_{CC}				
1 T_C = +25°C	I_{IH1}	3010	59	2.7 V									GND									5.5 V	5.5 V	T/R		20	µA
		"	60	5.5 V									"									2.7 V	"	$\overline{OE}$		20	"
	I_{IH2}	"	61	7.0 V									"									5.5 V	"	T/R		100	"
		"	62	5.5 V									"									7.0 V	"	$\overline{OE}$		100	"
	I_{IH3}	"	63	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.0 V	"	A0		1.0	mA
		"	64	"									"									"	"	A1		"	"
		"	65	"									"									"	"	A2		"	"
		"	66	"									"									"	"	A3		"	"
		"	67	"									"									"	"	A4		"	"
		"	68	"									"									"	"	A5		"	"
		"	69	"									"									"	"	A6		"	"
		"	70	"									"									"	"	A7		"	"
		"	71	0.0 V									"									"	"	B7		"	"
		"	72	"									"									"	"	B6		"	"
		"	73	"									"									"	"	B5		"	"
		"	74	"									"									"	"	B4		"	"
		"	75	"									"									"	"	B3		"	"
		"	76	"									"									"	"	B2		"	"
		"	77	"									"									"	"	B1		"	"
		"	78	"									"									"	"	B0		"	"
	I_{IL1}	"	79	"									"									0.5 V	"	$\overline{OE}$	1/	1/	"
	I_{IL2}	"	80	0.5 V									"									0.0 V	"	T/R	"	"	"

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	T/R	A0	A1	A2	A3	A4	A5	A6	A7	GND	B7	B6	B5	B4	B3	B2	B1	B0	$\overline{OE}$	V_{CC}				
T_C = +25°C	I_{OS}	3011	81	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	5.5 V	B0	-100	-325	mA
			82	"																				B1	"	"	
			83	"																				B2	"	"	
			84	"																				B3	"	"	
			85	"																				B4	"	"	
			86	"																				B5	"	"	
			87	"																				B6	"	"	
			88	"																				B7	"	"	
	I_{OSI}	"	89	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	"	A0	-60	-150	"
			90	"																				A1	"	"	
			91	"																				A2	"	"	
			92	"																				A3	"	"	
			93	"																				A4	"	"	
			94	"																				A5	"	"	
			95	"																				A6	"	"	
			96	"																				A7	"	"	
	I_{IOZL}		97	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	2.0 V	5.5 V	B0	1/	1/	"
			98	"																				B1	"	"	
			99	"																				B2	"	"	
			100	"																				B3	"	"	
			101	"																				B4	"	"	
			102	"																				B5	"	"	
			103	"																				B6	"	"	
			104	"																				B7	"	"	
			105	0.0 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	5.5 V	"	A0	"	"	"
			106	"																				A1	"	"	
			107	"																				A2	"	"	
			108	"																				A3	"	"	
			109	"																				A4	"	"	
			110	"																				A5	"	"	
			111	"																				A6	"	"	
			112	"																				A7	"	"	
	I_{IOZH}		113	"	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	A0		70	μA
			114	"																				A1		"	
			115	"																				A2		"	
			116	"																				A3		"	
			117	"																				A4		"	
			118	"																				A5		"	
			119	"																				A6		"	
			120	"																				A7		"	

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 20260-21T14:25Z7
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				T/R	A0	A1	A2	A3	A4	A5	A6	A7	GND	B7	B6	B5	B4	B3	B2	B1	B0	$\overline{OE}$	V_{CC}				
1 T_C = +25°C	I_{OZH}		121 122 123 124 125 126 127 128	5.5 V " " " " " " "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	GND " " " " " " "	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.0 V " " " " " " "	5.5 V " " " " " " "	B0 B1 B2 B3 B4 B5 B6 B7		70 " " " " " " "	µA " " " " " " "
	I_{CCH}	3005	129 130	" 0.0 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	" "	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.0 V "	" "	V_{CC}		110 110	mA "
	I_{CCL}	"	131 132	5.5 V 0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" "	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	" "	" "	"		130 130	" "
	I_{CCZ}	"	133 134	5.5 V 0.0 V									"									5.5 V 5.5 V	" "	"		143 143	" "
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																										
3	Same tests, terminal conditions, and limits as for subgroup 2, except T_C = -55°C and V_{IC} tests are omitted.																										
7 T_C = +25°C	Functional tests 2/	3014	135 136 137 138 139 140 141 142	B " " " A " " "	L H H L B A A B	L H L H B A B A	L H H L B A A B	L H L H B A B A	L H L H B A B A	L H L H B A B A	L H L H B A B A	L H L H B A B A	GND " " " " " " "	B A B A L H L H	B A B A B L H L	B A B A L H L H	B A B A L H H L	B A B A L H L H	B A B A L H L H	B A B A L H L H	B A B A L H H L	B A B A L H H L	B " " " " " " "	3/ " " " " " " "	All outputs " " " " " " "			
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																										
9 T_C = +25°C	t_{PLH1}	3003	143 144 145 146 147 148 149 150	0.0 V " " " " " " "	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	0.0 V " " " " " " "	5.0 V " " " " " " "	A0 to B0 A1 to B1 A2 to B2 A3 to B3 A4 to B4 A5 to B5 A6 to B6 A7 to B7	1.5 " " " " " " "	6.0 " " " " " " "	ns " " " " " " "
	t_{PLH2}	"	151 152 153 154 155 156 157 158	2.7 V " " " " " " "	IN	IN	IN	IN	IN	IN	IN	IN	" " " " " " " "	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	" " " " " " " "	" " " " " " " "	A0 to B0 A1 to B1 A2 to B2 A3 to B3 A4 to B4 A5 to B5 A6 to B6 A7 to B7	" " " " " " " "	" " " " " " " "	" " " " " " " "

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 20260210-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R_s,S,2 Test no.	1 T/R	2 A0	3 A1	4 A2	5 A3	6 A4	7 A5	8 A6	9 A7	10 GND	11 B7	12 B5	13 B5	14 B4	15 B3	16 B2	17 B1	18 B0	19 $\overline{OE}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003	159	0.0 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	IN	IN	IN	IN	IN	IN	IN	IN	0.0 V	5.0 V	B0 to A0	1.5	6.5	ns
		"	150	"									"											B1 to A1	"	"	"
		"	151	"									"											B2 to A2	"	"	"
		"	162	"									"											B3 to A3	"	"	"
		"	163	"									"											B4 to A4	"	"	"
		"	164	"									"											B5 to A5	"	"	"
		"	165	"									"											B6 to A6	"	"	"
		"	166	"									"											B7 to A7	"	"	"
	t_{PHL2}	"	167	2.7 V	IN	IN	IN	IN	IN	IN	IN	IN	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	A0 to B0	"	"	"
		"	168	"									"											A1 to B1	"	"	"
		"	169	"									"											A2 to B2	"	"	"
		"	170	"									"											A3 to B3	"	"	"
		"	171	"									"											A4 to B4	"	"	"
		"	172	"									"											A5 to B5	"	"	"
		"	173	"									"											A6 to B6	"	"	"
		"	174	"									"											A7 to B7	"	"	"
	t_{PLZ1}	"	175	0.0 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	-	"	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	IN	"	$\overline{OE}$ to A0	2.0	6.5	"
		"	176	"									"									"		$\overline{OE}$ to A1	"	"	"
		"	177	"									"									"		$\overline{OE}$ to A2	"	"	"
		"	178	"									"									"		$\overline{OE}$ to A3	"	"	"
		"	179	"									"									"		$\overline{OE}$ to A4	"	"	"
		"	180	"									"									"		$\overline{OE}$ to A5	"	"	"
		"	181	"									"									"		$\overline{OE}$ to A6	"	"	"
		"	182	"								OUT	"									"		$\overline{OE}$ to A7	"	"	"
	t_{PLZ2}	"	183	2.7 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	0.0 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	$\overline{OE}$ to B0	"	"	"
		"	184	"									"									"		$\overline{OE}$ to B1	"	"	"
		"	185	"									"									"		$\overline{OE}$ to B2	"	"	"
		"	186	"									"									"		$\overline{OE}$ to B3	"	"	"
		"	187	"									"									"		$\overline{OE}$ to B4	"	"	"
		"	188	"									"									"		$\overline{OE}$ to B5	"	"	"
		"	189	"									"									"		$\overline{OE}$ to B6	"	"	"
		"	190	"									"									"		$\overline{OE}$ to B7	"	"	"
	t_{PHZ1}	"	191	0.0 V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	"	$\overline{OE}$ to A0	"	6.5	"
		"	192	"									"									"		$\overline{OE}$ to A1	"	"	"
		"	193	"									"									"		$\overline{OE}$ to A2	"	"	"
		"	194	"									"									"		$\overline{OE}$ to A3	"	"	"
		"	195	"									"									"		$\overline{OE}$ to A4	"	"	"
		"	196	"									"									"		$\overline{OE}$ to A5	"	"	"
		"	197	"									"									"		$\overline{OE}$ to A6	"	"	"
		"	198	"									"									"		$\overline{OE}$ to A7	"	"	"
	t_{PHZ2}	"	199	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	"	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	"	"	$\overline{OE}$ to B0	"	"	"
		"	200	"									"									"		$\overline{OE}$ to B1	"	"	"
		"	201	"									"									"		$\overline{OE}$ to B2	"	"	"
		"	202	"									"									"		$\overline{OE}$ to B3	"	"	"
		"	203	"									"									"		$\overline{OE}$ to B4	"	"	"
		"	204	"									"									"		$\overline{OE}$ to B5	"	"	"
		"	205	"									"									"		$\overline{OE}$ to B6	"	"	"
		"	206	"									"									"		$\overline{OE}$ to B7	"	"	"

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	Measured terminal	Test limits Min	Test limits Max	Unit
				T/R	A0	A1	A2	A3	A4	A5	A6	A7	GND	B7	B6	B5	B4	B3	B2	B1	B0	$\overline{OE}$	V_{CC}				
9 T_C = +25°C	t_{PZL1}	3003	207	0.0 V	OUT	OUT	OUT	OUT					GND	0.0 V								IN	5.0 V	$\overline{OE}$ to A0	2.5	10.0	ns
			208	"									"									"	"	$\overline{OE}$ to A1	"	"	"
			209	"									"									"	"	$\overline{OE}$ to A2	"	"	"
			210	"									"									"	"	$\overline{OE}$ to A3	"	"	"
			211	"					OUT				"				0.0 V	0.0 V				"	"	$\overline{OE}$ to A4	"	"	"
			212	"						OUT			"			0.0 V						"	"	$\overline{OE}$ to A5	"	"	"
			213	"							OUT		"		0.0 V							"	"	$\overline{OE}$ to A6	"	"	"
			214	"								OUT	"	0.0 V								"	"	$\overline{OE}$ to A7	"	"	"
	t_{PZL2}	"	215	2.7 V	0.0 V	0.0 V	0.0 V	0.0 V					"									"	"	$\overline{OE}$ to B0	"	"	"
			216	"					0.0 V				"									"	"	$\overline{OE}$ to B1	"	"	"
			217	"						0.0 V			"									"	"	$\overline{OE}$ to B2	"	"	"
			218	"							0.0 V		"									"	"	$\overline{OE}$ to B3	"	"	"
			219	"								0.0 V	"				OUT					"	"	$\overline{OE}$ to B4	"	"	"
			220	"									"					OUT				"	"	$\overline{OE}$ to B5	"	"	"
			221	"									"						OUT			"	"	$\overline{OE}$ to B6	"	"	"
			222	"									"	OUT	OUT	OUT				OUT	OUT	"	"	$\overline{OE}$ to B7	"	"	"
	t_{PZH1}	"	223	0.0 V	OUT	OUT	OUT	OUT					"									"	"	$\overline{OE}$ to A0	"	7.5	"
			224	"					OUT				"									"	"	$\overline{OE}$ to A1	"	"	"
			225	"						OUT			"									"	"	$\overline{OE}$ to A2	"	"	"
			226	"							OUT		"									"	"	$\overline{OE}$ to A3	"	"	"
			227	"								OUT	"				2.7 V	2.7 V				"	"	$\overline{OE}$ to A4	"	"	"
			228	"									"					2.7 V				"	"	$\overline{OE}$ to A5	"	"	"
			229	"									"						2.7 V			"	"	$\overline{OE}$ to A6	"	"	"
			230	"									"	2.7 V	2.7 V	2.7 V				2.7 V	2.7 V	"	"	$\overline{OE}$ to A7	"	"	"
	t_{PZH2}	"	231	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V					"									"	"	$\overline{OE}$ to B0	"	"	"
			232	"					2.7 V				"									"	"	$\overline{OE}$ to B1	"	"	"
			233	"						2.7 V			"									"	"	$\overline{OE}$ to B2	"	"	"
			234	"							2.7 V		"									"	"	$\overline{OE}$ to B3	"	"	"
			235	"								2.7 V	"				2.7 V					"	"	$\overline{OE}$ to B4	"	"	"
			236	"									"					2.7 V				"	"	$\overline{OE}$ to B5	"	"	"
			237	"									"						2.7 V			"	"	$\overline{OE}$ to B6	"	"	"
			238	"									"	OUT	OUT	OUT				OUT	OUT	"	"	$\overline{OE}$ to B7	"	"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																							1.0	8.0	"
	t_{PLH2}																								"	8.5	"
	t_{PHL1}																								"	8.0	"
	t_{PHL2}																								"	8.5	"
	t_{PLZ1}																								2.0	10.0	"
	t_{PLZ2}																								"	10.0	"
	t_{PHZ1}																								"	9.0	"

See footnotes at end of device type 03.

Source: https://assist.dla.mil -- Downloaded: 2026/0-21T14:25Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases R,S,Z Test no.	1 T/R	2 A0	3 A1	4 A2	5 A3	6 A4	7 A5	8 A6	9 A7	10 GND	11 B7	12 B6	13 B5	14 B4	15 B3	16 B2	17 B1	18 B0	19 $\overline{OE}$	20 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
10	t_{PHZ2}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																							2.0	9.0	ns
	t_{PZL1}																								"	13.0	"
	t_{PZL2}																								"	13.0	"
	t_{PZH1}																								"	9.5	"
	t_{PZH2}																								"	9.5	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																										

1/ I_{IL} and I_{IOZL} limits shall be as follows:

Test	Min/Max limits in (mA) for circuit A Fairchild	B Motorola	C Signetics
I_{IOZL}	-.25/-.65	-.04/-1.2	-.03/-1.0
I_{IL1}	-.05/-1.2	-.04/-1.2	-.06/-1.6
I_{IL2}	-.05/-1.2	-.04/-1.2	-.09/-1.2

2/ A = 3.0 V minimum, B = 0.0 V or GND; H > 1.5 V, L < 1.5 V

3/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:25Z
Check the source to verify that this is the current version before use.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design application and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - -	Ground zero voltage potential.
V_{IN} - - - - - - - - - - - - - -	Voltage level at an input terminal.
I_{IN} - - - - - - - - - - - - - -	Current flowing into an input terminal.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F242
02	54F243
03	54F245

6.6 Manufacturers' designators. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Fairchild	Motorola	Signetics
01		X	
02		X	
03		X	

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - SH, OS, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962- 0753)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/348A	MICROCIRCUITS, DIGITAL, ADVANCED SCHOTTKY TTL, BUS TRANSCEIVERS WITH THREE-STATE OUTPUTS, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:25Z
Check the source to verify that this is the current version before use.